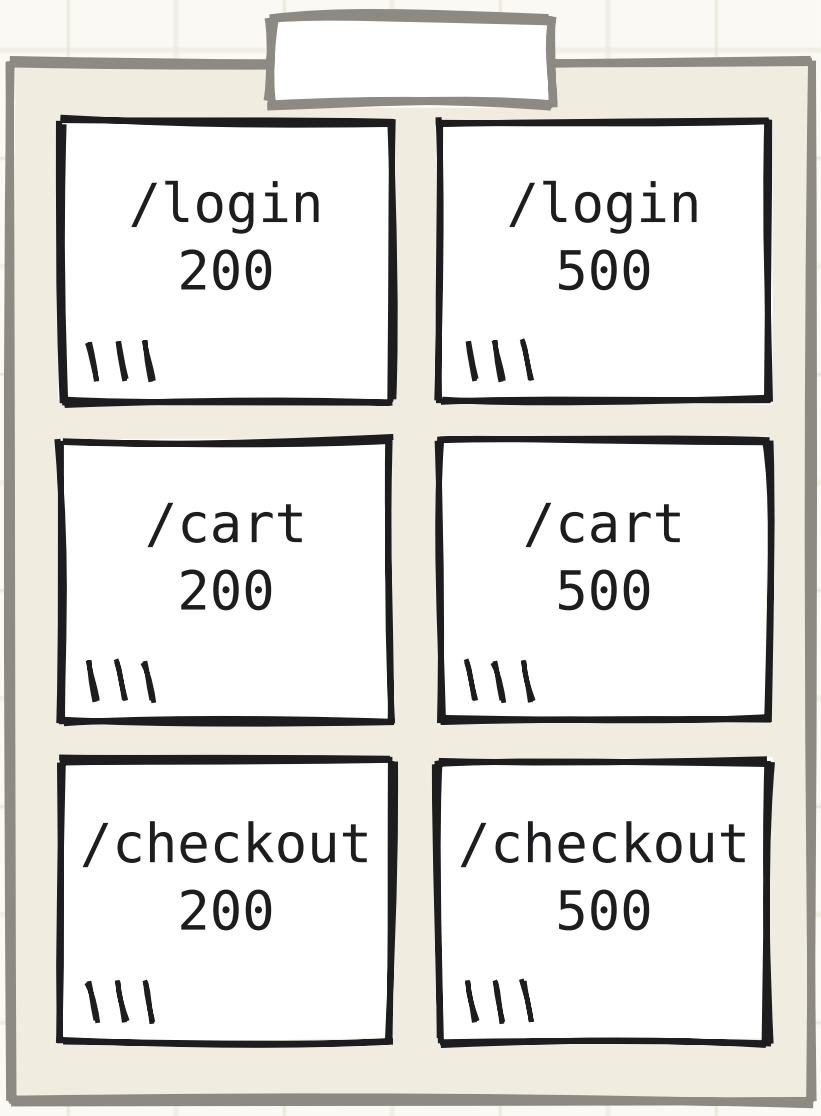
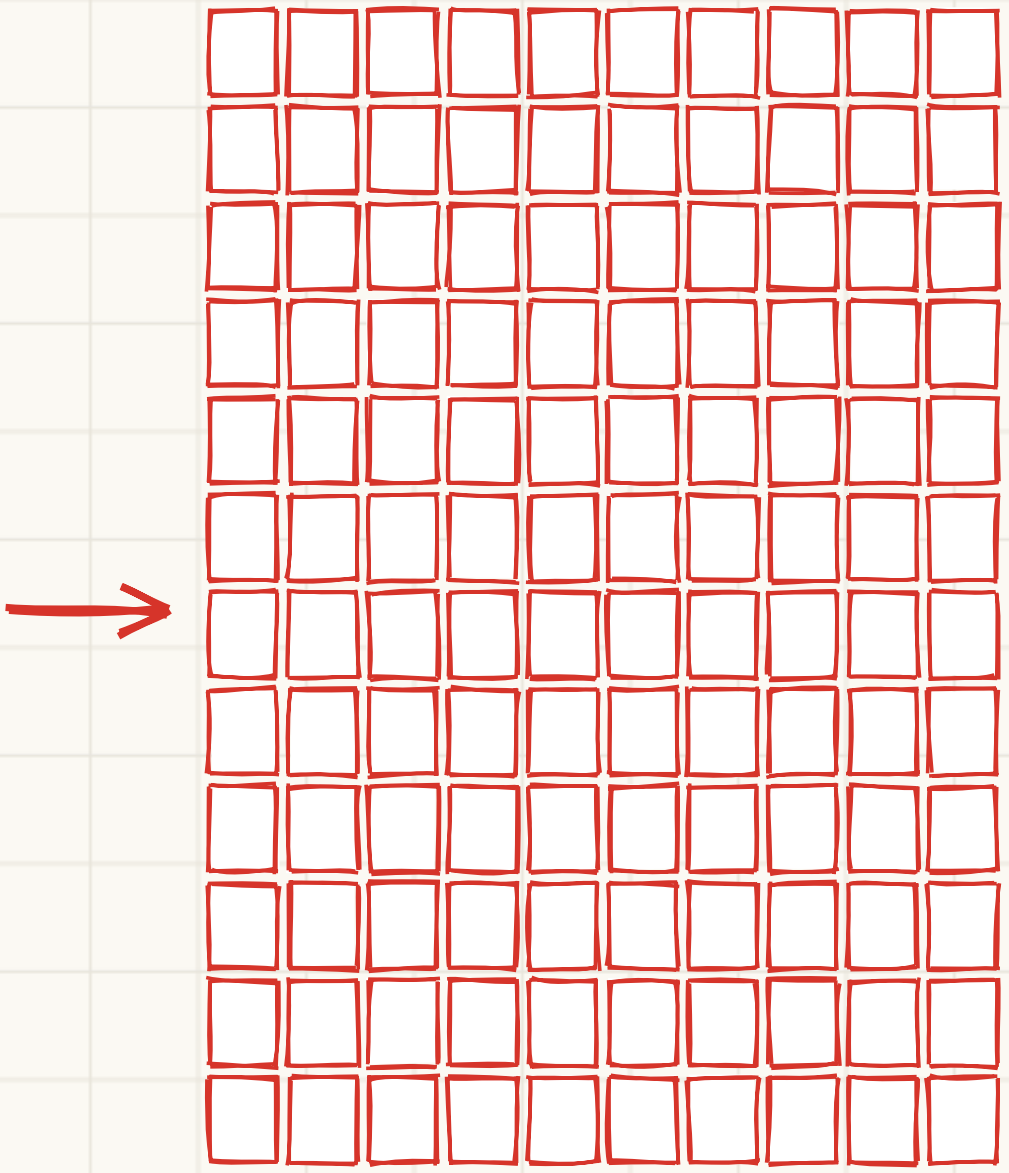
One label.
223,770 time series.

Prometheus keeps one tally sheet per label combo. I added `user_id`.



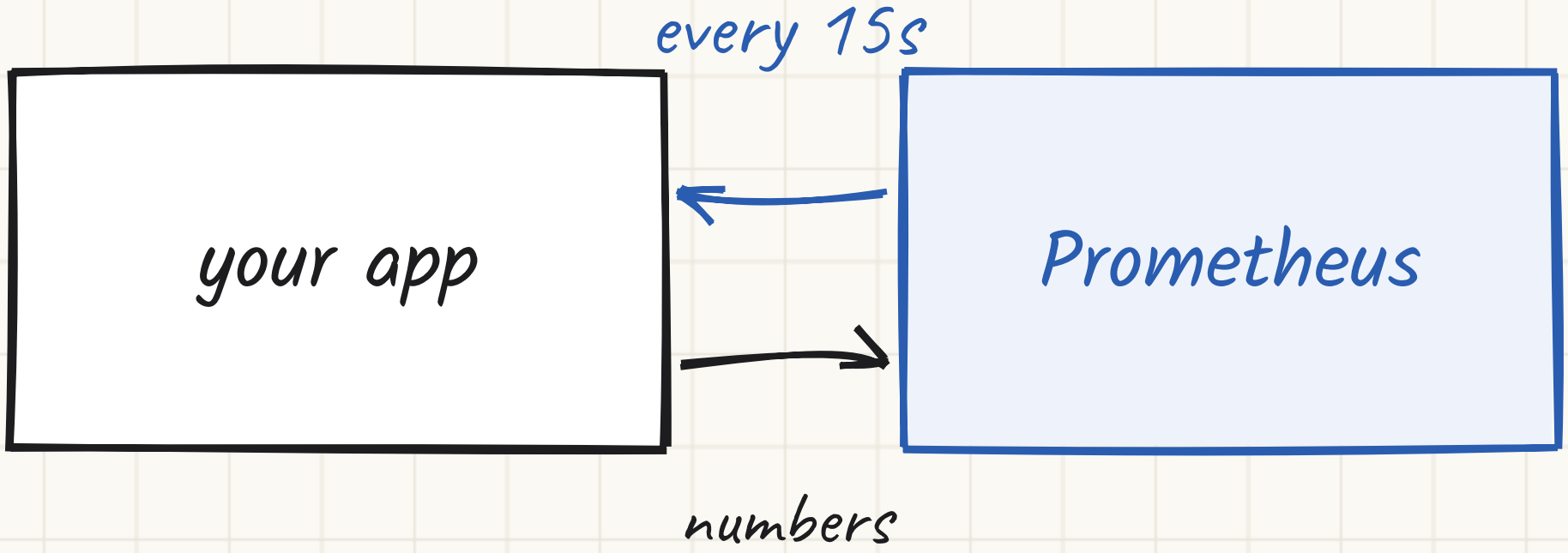
6 sheets



one per user

01 Prometheus writes down your counts.

Your app shows its numbers on a `/metrics` page. Every 15 seconds Prometheus reads it. That visit is a scrape.




```
GET /metrics
http_requests_total 3200
```

But which requests?

02 Labels are tags on a count.

A metric is a named number. Labels say which slice it counts: which page, which result.

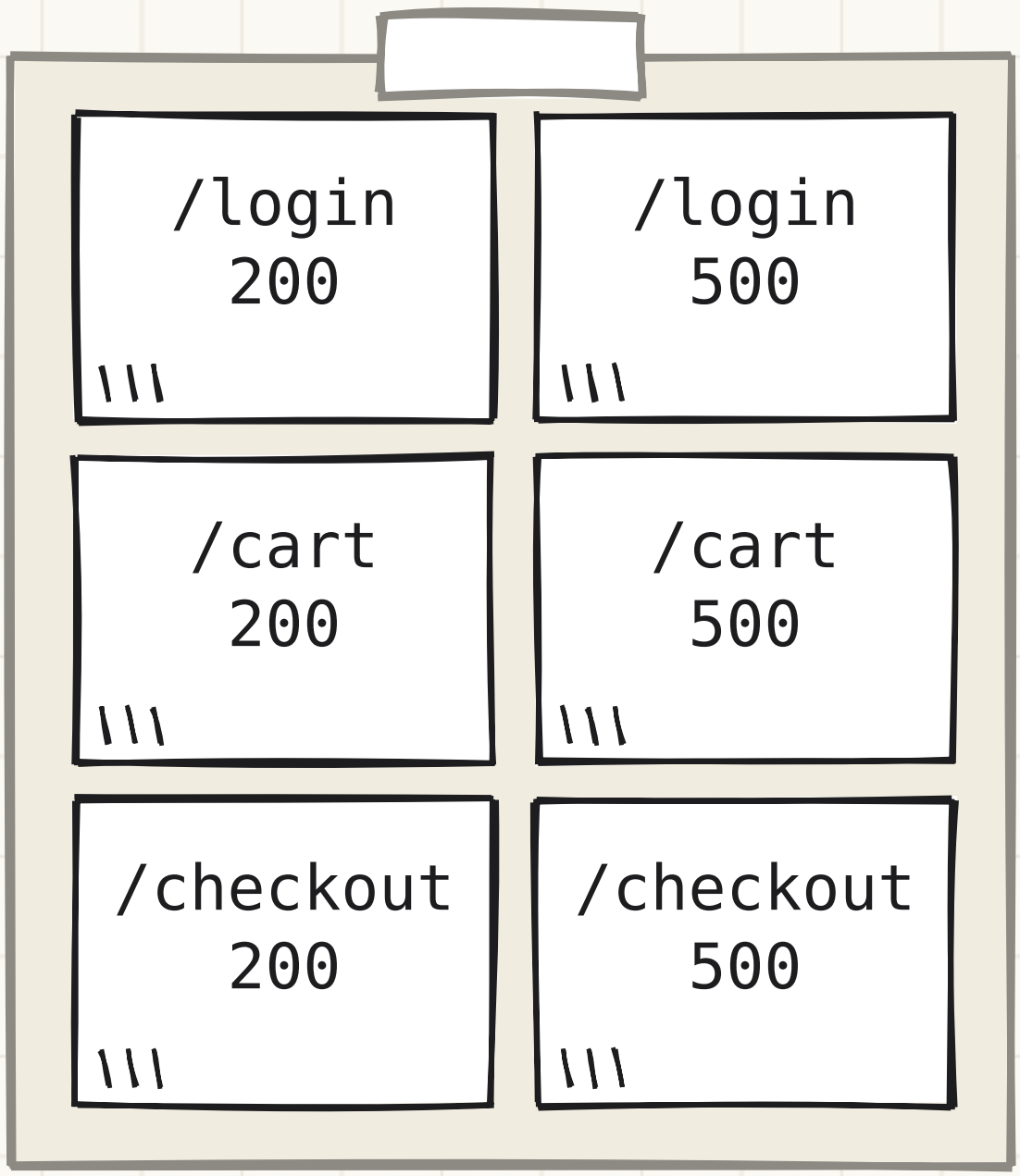
```
http_requests_total{
   route="/checkout",        label
   status="200"              label
} 849
```

the count

Now the key idea.

03 One sheet per tag combo.

3 routes x 2 results = 6 sheets. Each sheet is a time series: one number, ticked every scrape.



6 series

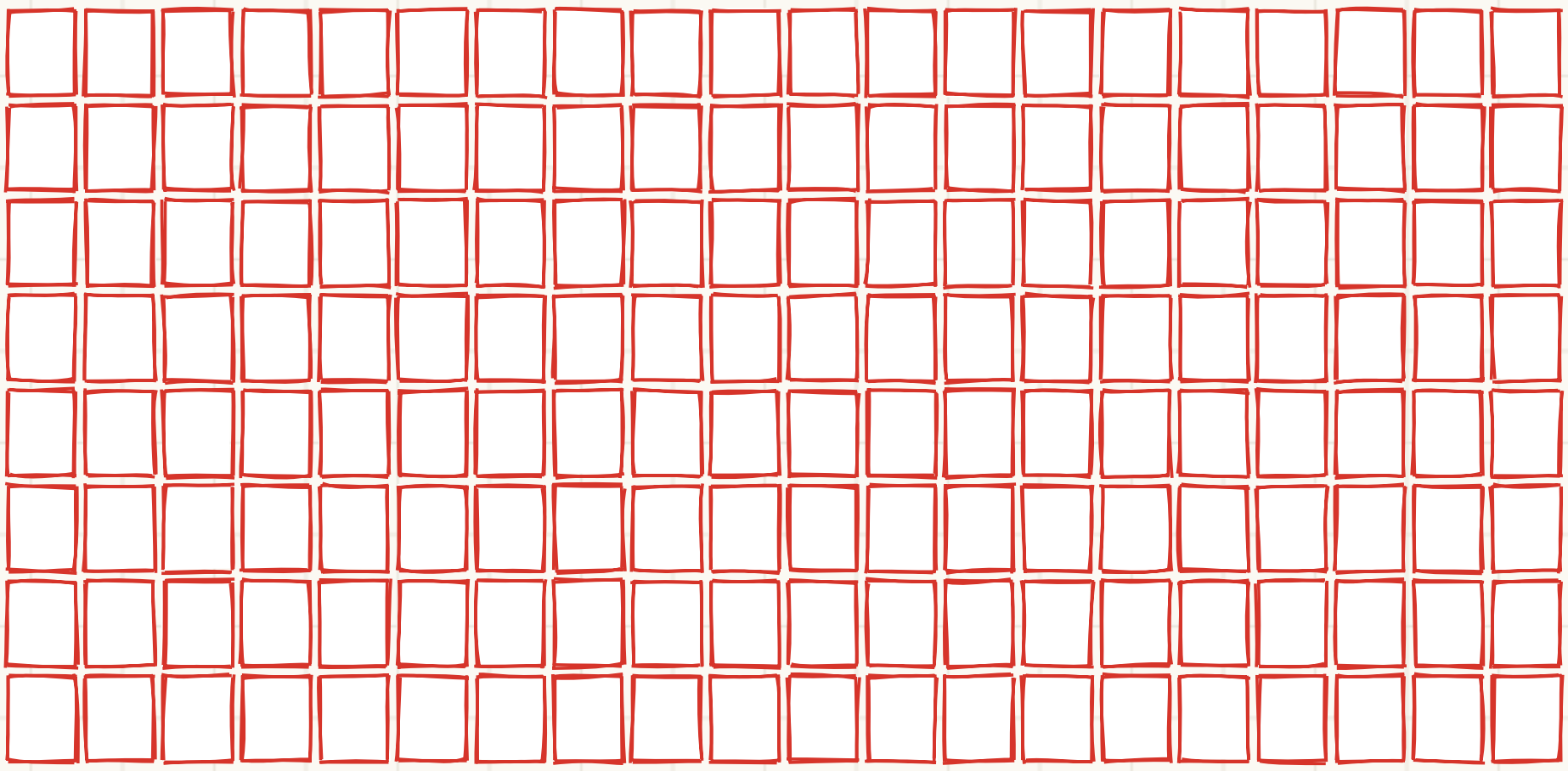
= 6 sheets

Then someone wanted per user data.

04 Wrong: add user_id as a label.

Every user gets their own sheets. My run with 10,000 users:

```
labels = ["route", "status", "user_id"]
```

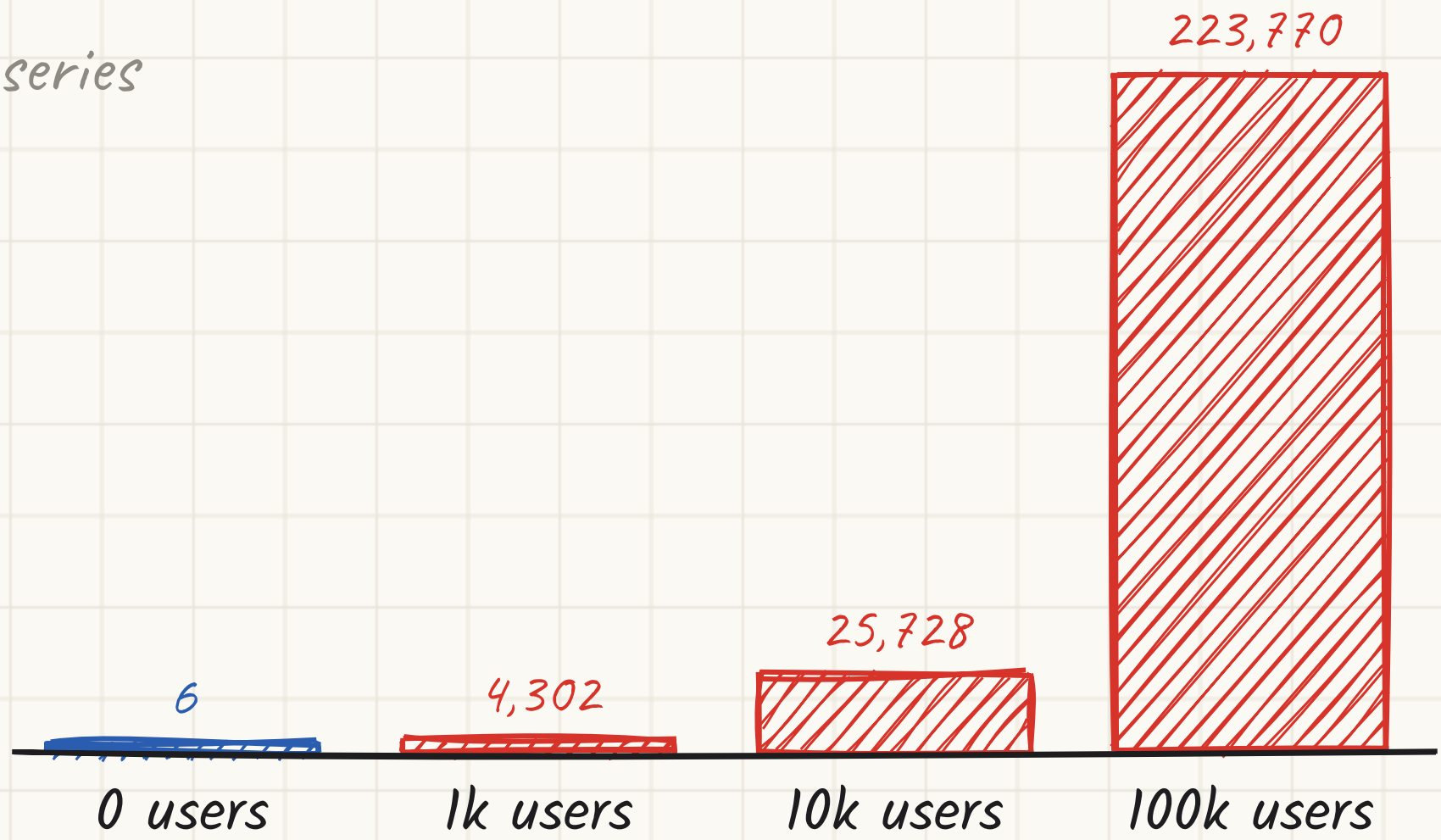


25,728 series

Sheets are not free.

05 Every sheet costs memory.

Same counter, more users. Real run, Prometheus 3.15.0, about 2.5 minutes of scrapes.



Prometheus memory

82 MB	98 MB	142 MB	571 MB

How do you spot it?

06 See it: count values per label.

Prometheus has a TSDB status API. It lists how many values each label has.

```
GET /api/v1/status/tsdb
```

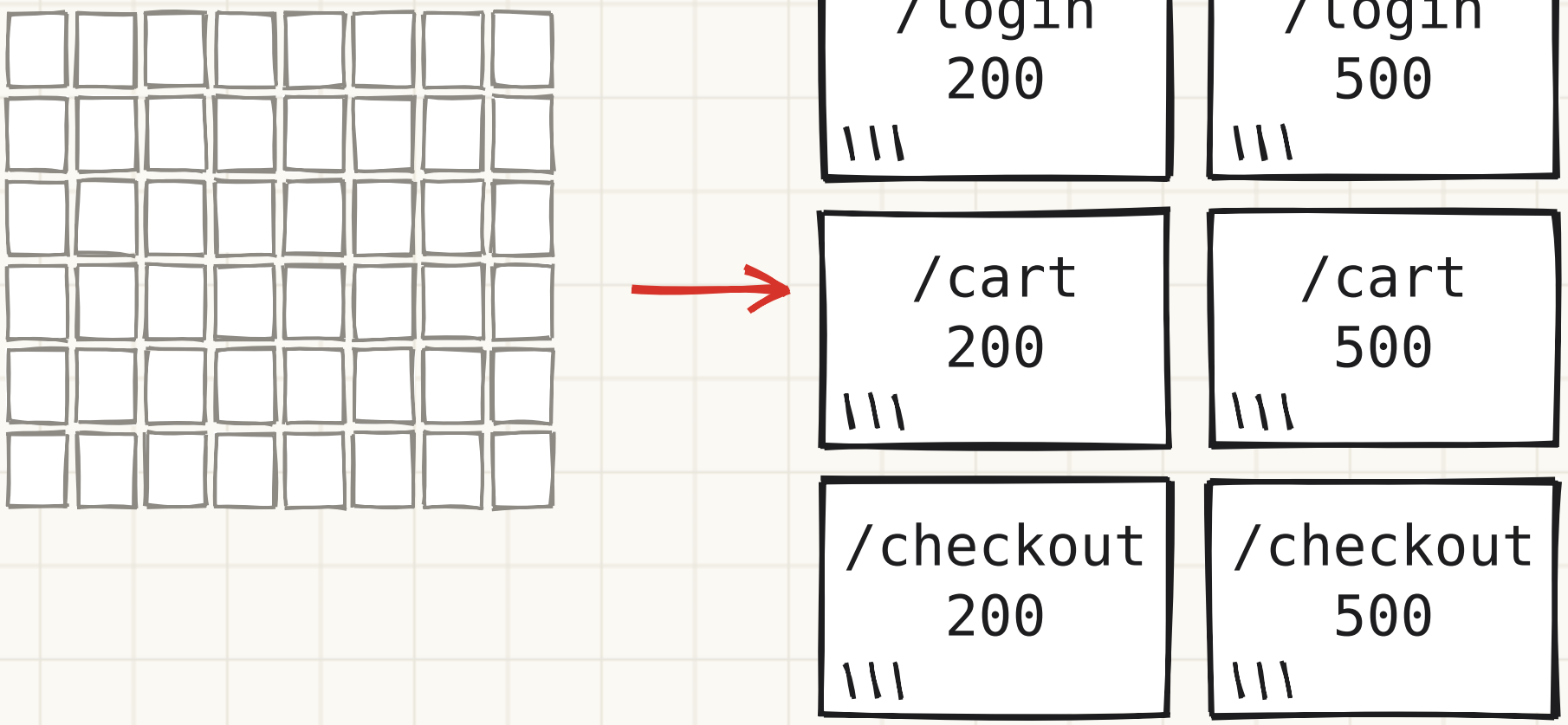
user_id	9794
__name__	6
route	3
status	2

the guilty label

Tempting quick fix...

07 Gotcha: deleting it at scrape lies.

`labeldrop user_id` gave 6 series again. But the total read 16, not 10,100. No error.



real total	10,100
after drop	16 ✕

So what is right?

08 Right: few sheets, plus a seatbelt.

Remove `user_id` in code. Per user detail goes to logs. Then cap each scrape.

1. `labels = ["route", "status"]`
   `-> 6 series` ✓

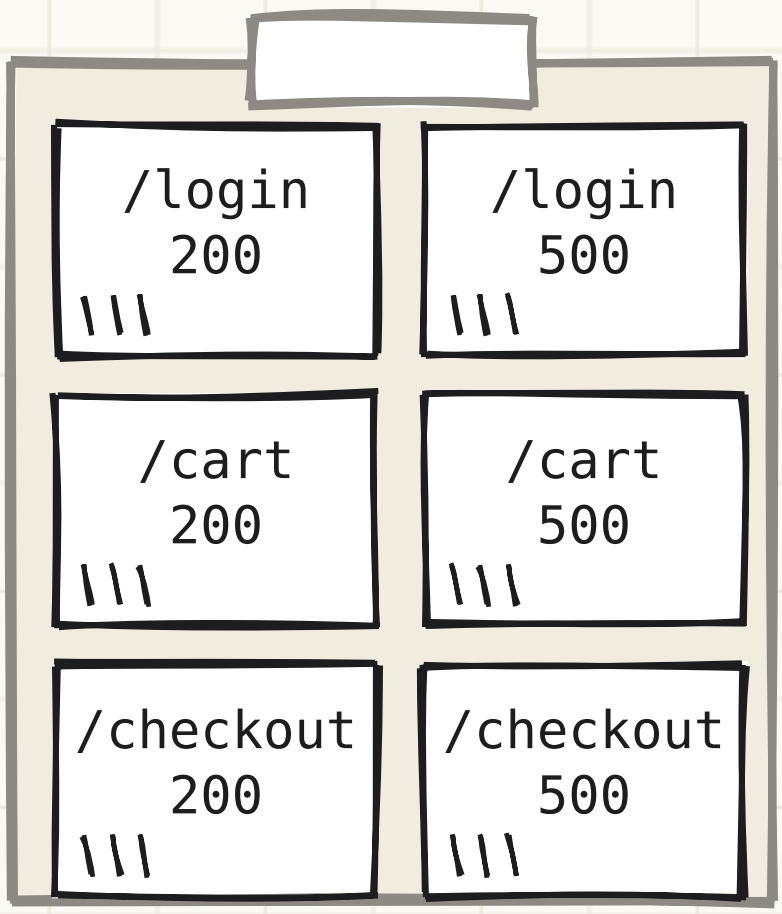
2. `sample_limit: 5000`
   `-> scrape refused, up = 0`

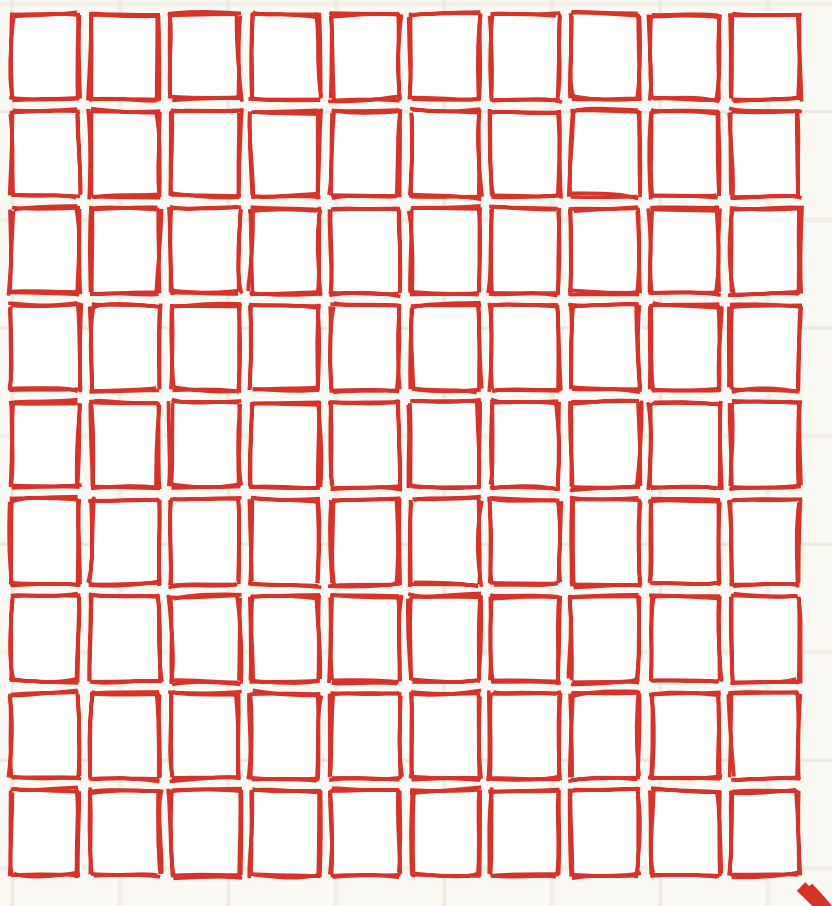
   `"sample limit exceeded"`

So, the whole drawing.

So, the whole drawing.

Every label combo is a sheet. Unbounded values mean unbounded sheets. Keep labels small, cap scrapes, put users in logs.





Which label in your metrics grows with your users?